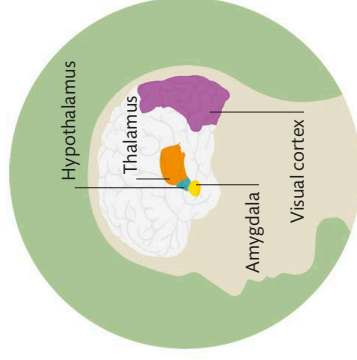


Fear and Anger

Fear and anger trigger the release of hormones in the body that prepare us to deal with threats. In the modern world, however, long-term anxiety can cause overactivation of the sympathetic nervous system and lead to health problems.

Fight or flight

When we see a possible threat, visual information travels to our amygdala, a tiny part of the brain that processes emotion. The amygdala sends a signal to the hypothalamus, which activates the sympathetic nervous system, preparing the body to react to danger (see p.13). The hypothalamus also sends signals to the pituitary and adrenal glands, which secrete hormones such as cortisol and adrenaline. The combined effect of these pathways is to initiate our fight-or-flight reflex, which prepares our bodies to attack or run away.



Responding to danger

Signals travel to the thalamus and amygdala, which triggers the hypothalamus to produce fight-or-flight hormones. A slower, conscious pathway involving the cortex also assesses the situation (see p.107).



Saliva production reduces

Saliva secretion slows down when we are afraid. This causes a dry mouth.



Pupils dilate

Our pupils enlarge, letting in more light so we can see the threat more clearly.



Muscles tense

The muscles in our arms, legs, and shoulders prepare themselves for action. We may feel tense or "wound up."



Digestion slows

To avoid wasting energy, digestive activity falls. In extreme cases, we may vomit to eject undigested food.



Breathing rate rises

This oxygenates our muscles, preparing them for action. But it can also cause symptoms of hyperventilation.



Heart rate increases

Our heart beats faster to pump oxygen-and-nutrient-rich blood to where it is needed in the body.



Sweating increases

Our sweat glands are triggered, and we begin to sweat, ensuring we remain cool if physical exertion is needed.



Blood vessels constrict

Blood flow is directed away from the surface of the skin, so we may appear pale.